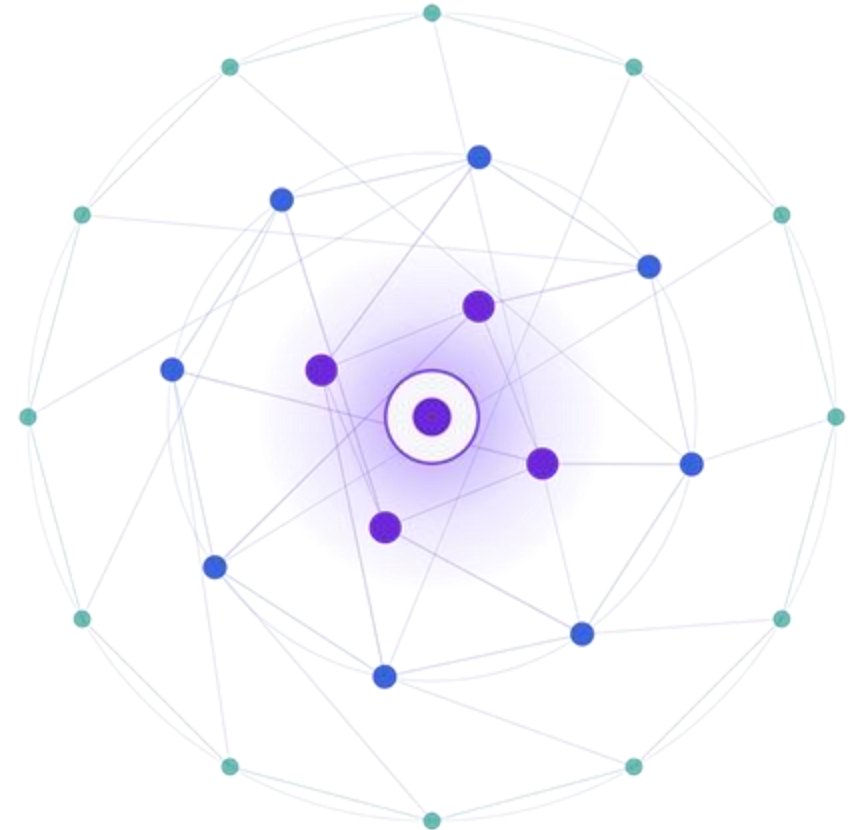


Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 05 (Interruptible Real-Time Agents)
- Team Name - BYTEbuzz
- College Name – SRM Institute of Science and Technology, Kattankulathur
- Member Name & Email 1- Palak Lohia (pl3571@srmist.edu.in)
- Member Name & Email 2- Srishti Srivastava (ss0681@srmist.edu.in)
- Member Name & Email 3- Akshata Srivastava (as1282@srmist.edu.in)
- Member Name & Email 4- Manashvi Sharma (ms7215@srmist.edu.in)
- Submission Github link – <https://github.com/srishti-1935/duplex-agent>



Theme

Build a voice-native agent, any architecture, inside the LiveKit agents framework, and evaluate it on **Full-Duplex-Bench v3**, a public benchmark from NTU with NVIDIA advisory.

1

Stays responsive

Spoken feedback within a few hundred ms. No dead air and no false “done” claims.

2

Works asynchronously

Tool calls, perception and reasoning never block the conversation.

3

Recovers cleanly

Mid-utterance corrections discard stale intent, update tool arguments, and never repeat a state-changing action.

100

real human recordings

79

unique scenarios, 12 speakers

5

disfluency types, incl. self-corrections

12

mock tools, 4 domains, 1–3 chained calls

Refund order 4... no **wait**,
5... uh, the blue one

100

•

79

•

12

Existing Solutions & Gaps

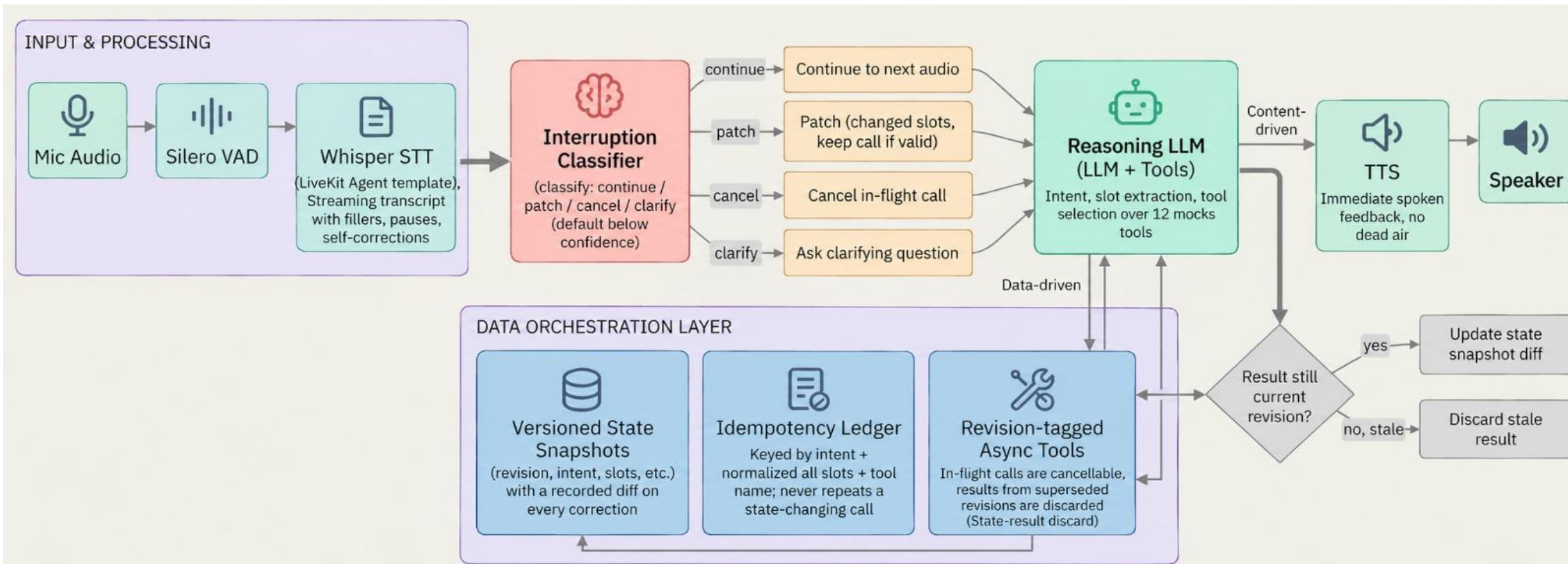
Even the best systems fail corrections

a table using the published baselines from the benchmark paper.

System	Strict pass	Task-completion latency
GPT-Realtime	60.0%	6.89s
Gemini Live 3.1	54.0%	4.25s
Cascaded (Whisper→GPT-4o→TTS)	n/a	10.12s

The paper's own finding: self-corrections and multi-step tool chains are where everyone loses points.

Our Solutions & Architecture Diagram



Why cascaded: clean STT → reasoning → TTS separation lets us insert classification and state management between them.

Safety rule: low-confidence interruptions default to clarify, so the agent never guesses on a state-changing action.

Demo & Product Walkthrough

 **INTERRUPT.AI**
AGENT WORKSPACE

WORKSPACE

Trace Studio

Scenarios

Tool Ledger

Trace Studio > Live Session

SIMULATOR READY

DEMO MODE

Reset session

LIVE EXECUTION

Trace Studio

Observe how the agent processes events, handles interruptions, and protects state in real time.

Run scenario

Reset

INTERRUPTIONS

01

STATE REVISION

v2

TOOL CALLS

02

PROTECTED

01

SCENARIO RUNNER

DATE CORRECTION

Replay an interruption sequence and inspect the resulting state.

DATE CORRECTION

Run scenario

EVENT STREAM

6 events

15:11:03.729	TEXT	Find me a flight to Mumbai on October 4. revision v1 - received
15:11:04.077	TOOL_CALL	flight_search revision v1 - received
15:11:03.528	INTERRUPTION	Actually make that October 5. revision v1 - received
15:11:03.777	DECISION	PATCH revision v1 - received
15:11:04.078	STATE_CHANGE	date: October 4 → October 5 revision v2 - received
15:11:03.627	TOOL_RESULT	Result accepted revision v2 - received

INTERRUPTION DECISION

PATCH

CONFIDENCE

98%

CHANGED SLOTS

date

ACTION

Update only the changed slot.

Made in Bolt

User: “Book a table for four at seven... no wait, make it eight.”

Agent: “Sure, moving it to eight.”

Classifier: patch: Revision 1 → 2, slot diff: time 7 PM → 8 PM

Ledger check: No duplicate booking; stale result discarded

Tools and tech stack used

- **LIVEKIT AGENTS
SPEECH**

Silero · Whisper · OpenAI TTS

- **GPT-4o**

Function Calling

- **BENCHMARK**

Full-Duplex-Bench v3

- **PYTHON 3.12**

- **REPRODUCIBLE**

One-command · pinned versions



One command reproduces our benchmark run.

Impact & Use case

From benchmark to real life: Hands-Free Navigation



Hands-free situations where users constantly correct themselves.

THE PROBLEM

Changing your destination while driving means taking your eyes/hands away from the road.

THE KEY MOMENT

User changes their mind mid-sentence.
The agent doesn't restart.
It updates the action.

Innovation highlights, results and limitations

What worked, and what didn't

**Understands
mid-sentence
changes**

CONTINUE

PATCH

CANCEL

CLARIFY

**Keeps the
latest intent**

- Revision tracking
- Prevents duplicate tool calls
- Rejects outdated results

**Control comes
with cost**

- Cascaded pipeline adds latency
- CPU setup affects response time
- Native realtime integration had issues

Interrupt → Understand → Update → Act

What's next

01 — Expand Use Cases

Extend interruption-aware agents to:

- In-car assistants
- Smart-home control
- Hands-free productivity
- Customer support

02 — Multimodal Interruptions

Go beyond voice:

- Camera/image context
- Screen interaction
- Visual corrections
- Real-time multimodal inputs

03 — Improve Responsiveness

- Reduce first-response latency
- Improve cancellation speed
- Handle longer tool chains
- Improve recovery from complex corrections

04 — Smarter Adaptive Agents

- Learn user interaction patterns
- Better intent recovery
- More reliable multi-step planning
- Safer autonomous tool execution

Vision: An AI assistant that can think, act, listen and recover — without losing the user's intent.

Brownie points slide (differentiation)

1

Four-way interruption decisions

Every correction is classified as continue, patch, cancel or clarify before any action. Low confidence defaults to clarify.

2

No repeated state-changing action

An idempotency ledger keyed by intent, normalized slots and tool name blocks duplicate calls.

3

Stale results never surface

Tool calls carry a revision tag. Superseded results are dropped on arrival and in-flight calls are cancelled.

4

Reproducible and honest

One-command run, pinned seeds and versions, no caching across scenarios, and no hardcoding or fine-tuning on FDB-v3.

Aimed at the paper's own headline finding: self-correction and multi-step tool chains are where every published system loses points.

Checklist- Updated on Public GitHub



Working prototype code

Public or shared GitHub repo: duplex-agent

Y · Attached



README with reproducible setup instructions

Architecture diagram, exact setup and run steps, extension clearly marked

Y · Attached



Demo video, max 5 minutes

Real benchmark interruption, then the extension use case

Y · Attached



Presentation file (PPT or PDF)

This deck, submitted as a PPT via the Google Form

Y · Attached

Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India